

Onkaar Grewal

Onkaar.Grewal@outlook.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | Cambridge, ON

TECHNICAL SKILLS

Design & CAD: SolidWorks, Autodesk Inventor, AutoCAD, Fusion 360, KiCad, Blender

Programming & Tools: Python, C/C++, MATLAB, Arduino, LaTeX, Granta EduPack

Hardware & Fabrication: Embedded Systems, 3D Printing (FDM, TPU/PLA), Soldering, Raspberry Pi, Power Tools

PROJECTS

Portable Weather Station Node | *Arduino, C/C++, Sensors, SolidWorks* 2026 – Present

- Engineered an autonomous weather monitoring system with a sensor suite measuring **humidity, temperature, wind speed, and rainfall** with **real-time data logging**.
- Designed and fabricated a **mechanical housing in SolidWorks** to protect electronics from environmental exposure while maintaining accurate sensor readings.
- Implemented **sensor communication protocols** and **data logging pipelines** on an **Arduino-based embedded system**.

Power in Community (AeroPup) – Assistive Ball Launcher | *Autodesk Inventor, TPU/PLA 3D Printing, Arduino* 2026

- Designed and built a **wheelchair-mounted ball launcher** achieving a **5m+ range** within a **sub-1kg** weight limit for a client with **Juvenile Rheumatoid Arthritis**, serving as **Materials Subject Matter Expert** and mechanical designer.
- Selected **TPU (gyroid infill)** for custom-grooved tires and **PLA (rigid infill)** for structural components, balancing grip and energy transfer against weight and stiffness constraints.
- Engineered **custom motor-dampening walls** and a **servo-driven rack-and-pinion elevation system**, reducing noise and vibration to public-use levels across four design iterations.

Robotic End Effector (Sir Lifts-A-Lot) | *Autodesk Inventor, Python, Q-Arm* 2025

- Designed a **robotic end effector** for a Q-Arm to automate packaging of **irregularly shaped warehouse items**, serving as **Project Manager** and mechanical designer for a 4-person team.
- Engineered a **sponge-and-rubber-foam claw** for surface-area grip, re-angling the assembly with a **50-degree offset** to match the Q-Arm's kinematic constraints.
- Wrote a **Python barcode-lookup function** cross-referencing a product database to drive the arm's sorting logic; final design handled **5 of 5** target objects successfully.

EXPERIENCE

Engineering Design & Drafting Intern Cambridge, ON
Atlantic Industries Limited January 2023 – June 2023

- Produced **high-accuracy 2D AutoCAD drawings** for **structural civil engineering projects**, transitioning from administrative support to technical drafting.
- Collaborated with university co-op students on **large-scale infrastructure designs** and assisted in **document management** and drawing revisions.

First Robotics Competition Mentor Cambridge, ON
St. Benedict Catholic Secondary School September 2023 – January 2025

- Mentored **mechanical sub-teams** in **CAD prototyping and fabrication** of competitive **120lb robots**, guiding **drivetrain and intake design** layout.
- Instructed junior students on **shop safety** and safe operation of **power tools and industrial machinery**.

Administrative Project Support Ontario, Canada
YMCA of Three Rivers March 2025 – Present

- Coordinate **board-level initiatives and events** across **6 municipal locations**, managing **project documentation** and scheduling for regional leadership.
- Facilitate **cross-location communication** for the **Board of Directors** to ensure efficient execution of community-focused projects.

EDUCATION

McMaster University Hamilton, ON
Materials Science and Engineering (Co-op) - Year 2 Sep 2025 – Present